

tf.pad crashes with large paddings #51908

✓ Closed

lugalUrim opened on Sep 10, 2021

System information

- Have I written custom code (as opposed to using a stock example script provided in TensorFlow): Yes
- OS Platform and Distribution (e.g., Linux Ubuntu 16.04): Linux Ubuntu 18.04
- Mobile device (e.g. iPhone 8, Pixel 2, Samsung Galaxy) if the issue happens on mobile device: N/A
- TensorFlow installed from (source or binary): binary
- TensorFlow version (use command below): 2.6.0
- Python version: 3.6.8
- Bazel version (if compiling from source): N/A
- GCC/Compiler version (if compiling from source): N/A
- CUDA/cuDNN version: N/A
- GPU model and memory: N/A

Describe the current behavior

`tf.pad` crashes when the argument "paddings" has large values.

Describe the expected behavior

Expect an exception to be thrown if the input `paddings` is unexpected.

Standalone code to reproduce the issue

```
import tensorflow as tf
input_tensor = tf.random.uniform([1, 32, 32, 3], dtype=tf.float32)
paddings = [[125106557, 1415887920], [747509374, 2136925906], [413308538, 904601717],
[1900762018, 831358864]]
res = tf.pad(input_tensor, paddings)
```

outputs:

```
2021-09-09 12:46:38.123113: F tensorflow/core/framework/tensor_shape.cc:352] Check
failed: 0 <= new_num_elements (0 vs. -1)
Aborted (core dumped)
```

 **lugalUrim** added **type:bug** on Sep 10, 2021

 **google-ml-butler** assigned sushreebarsa on Sep 10, 2021

 **sushreebarsa** added **2.6.0** **comp:ops** on Sep 10, 2021

 sushreebarsa on Sep 10, 2021

Contributor ...

@lugalUrim Could you please have a look at the [link](#), and similar [issue](#) ,Please let us know if it helps ?
Thanks!

 **sushreebarsa** added **stat:awaiting resp...** on Sep 10, 2021

 lugalUrim on Sep 11, 2021

Author ...

Yes, thanks **@sushreebarsa** for pointing to that. I will close this issue.

 **lugalUrim** closed this as completed on Sep 11, 2021

 google-ml-butler **bot** on Sep 11, 2021

...

Are you satisfied with the resolution of your issue?

[Yes](#)

[No](#)

 **yongtang** mentioned this on Sep 13, 2021

 Fix tf.pad crashes with large paddings #51973

 yongtang on Sep 13, 2021

Member ...

In case of check failure, tensorflow should return an error gracefully instead of crash with undefined behaviors. Created a PR [#51973](#) for the fix.

  **rasabot** mentioned this on Nov 13, 2021

 Security Alert RasaHQ/rasa#8638

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Assignees

Labels

2.6.0

comp:ops

stat:awaiting response

type:bug

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

 Code with agent mode

▼

 **Fix tf.pad crashes with large paddings**

tensorflow/tensorflow

Participants

